# Exercise sheet II

**LV 4502 – Advanced Macroeconometrics**

Due date: Friday, 02.07.2021

## 1 VAR Coding (10pts)

Remember a VAR(p) model

$$y_t = \sum_{j=1}^{p} A_p y_{t-j} + c + \varepsilon_t$$

where we implemented the independent NiW-prior with the following structure for the coefficients

$$p(A) \sim \mathcal{N}(\underline{A}, \underline{V}),$$
$$p(\Sigma) \sim iW(\underline{s}, \underline{S})$$

with prior expectation and variance

$$\underline{A} = E[A] = (I_M, 0, \ldots, 0)$$
$$\underline{V} = Var(A) = \begin{cases} \left(\frac{\lambda_1}{k}\right)^2 & \text{for } i = j \text{ and the } k\text{-th lag and,} \\ \left(\frac{\sigma_i^2}{\sigma_j^2}\right)\left(\frac{\lambda_2}{k}\right)^2 & \text{for } i \neq j \text{ and the } k\text{-th lag,} \\ \lambda_3 \sigma_i^2 & \text{for the deterministic part of the model,} \end{cases}$$

and set the hyperparameter for the inverse-Wishart prior accordingly. Your task is to build a hyperprior for the hyperparameters $\lambda_1$, $\lambda_2$ and $\lambda_3$. Assume that

$$\lambda_j \sim G(0.01, 0.01) \quad \forall j$$

and implement a RW-MH step to the existing sampler. Within this setting, please re-estimate the three-variable monetary policy model discussed in class. Compare the results and discuss. Note that you have to specify an *asymmetric* proposal density due to the fact that the support of the Gamma distribution is $R^+$. Hence, as opposed to the simplified RW-MH with symetric proposal density discussed in class, you have to take into account the effect of an asymetric

proposal density[1]. To do this, use the following acceptance probability:

$$\min\left[1, \frac{f(\theta^{prop})\alpha(\theta^c\theta^{prop})}{f(\theta^c)\alpha(\theta^{prop}\theta^c)}\right]$$

with $f(\cdot)$ denoting the posterior density, $\alpha(\cdot)$ denoting the proposal density, $\theta^c$ denoting the current parameter value and $\theta^{prop}$ denoting the proposed parameter value. Consider drawing the proposal from LogNormal distribution

$$\log(x) \sim N(x^c, \kappa),$$

and tune $\kappa$ accordingly.

**Relevant Literature:**


- **Kilian, L. and H. Lütkepohl (2017):** Structural Vector Autoregressive Analysis. Cambridge University Press.
- **Lynch, S. M. (2007):** Introduction to Applied Bayesian Statistics and Estimation for Social Scientists. Springer Science+Business Media.


[1] see also Chapter 5 of the Lynch (2007) textbook